AMERICAN PSYCHOLOGICAL ASSOCIATION

Emotion

ISSN: 1528-3542

2026, Vol. 26, No. 3, 771–786
https://doi.org/10.1037/emo0001614

Visual Attention to Emotional Pictures: Striking Parallels With Neutral Stimuli Challenge Emotion-Specific Accounts of Influences on Attentional Biases

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Converging evidence suggests that visual spatial attention is preferentially allocated to emotional over neutral stimuli, referred to as an attentional bias to emotional information. Intriguing questions emerged about whether this attentional bias is facilitated by an assumed right hemispheric dominance of emotion processing and by converging cross-modal information (Gerdes et al., 2021). However, we argue that a critical condition that would allow an interpretation in terms of an influence on an emotional attention bias is missing from the experimental design testing these effects: namely, a condition presenting only neutral pictures. To corroborate our argument, we conducted a replication and extension of the eye-tracking study by Gerdes et al. (2021), including this control condition. Specifically, we presented pairs of pictures and lateralized sounds in a free-viewing paradigm and tested the effect of picture position, sound position, and sound valence on an attentional bias score (BS), a difference value for the number of first fixations to unpleasant pictures compared to neutral ones. Importantly, we included a neutral–neutral condition and computed a corresponding BS for an arbitrary set of neutral pictures. Both the supposed leftward bias of emotional attention (i.e., the BS for unpleasant pictures is more pronounced if they are presented on the left) and its supposed guidance by sounds (i.e., the BS for unpleasant pictures is more pronounced if a sound was heard on the same side as the unpleasant picture) emerged in striking parallelism for both unpleasant–neutral and neutral–neutral picture pairs. Thus, the paradigm gives no evidence for emotion specificity of results.

Keywords: cross-modal, eye-tracking, attention, emotion, bias score

Supplemental materials: https://doi.org/10.1037/emo0001614.supp

In natural environments, the human cognitive system is constantly confronted with an abundance of information from different senses, leading to competition for attentional selection within and across sensory modalities. A large body of research has addressed the question of whether certain classes of meaningful stimuli, such as those with emotional value, have an advantage in this competition for selective attention. For theoretical reasons (e.g., Lang et al., 2000; Öhman, 2005), this research has initially focused on the attentional advantage for negative stimuli and, for theoretical as well as pragmatic reasons, predominantly on visual attentional biases (e.g., Yiend, 2010). In the visual modality, several lines of evidence have demonstrated a preferential allocation of attention to emotional over neutral stimuli in the general population (for reviews, see, e.g., Pourtois et al., 2013; Yiend, 2010). This is often referred to as an attentional bias to emotional, or particularly to negative, information (e.g., Yiend, 2010). However, emotional information may contribute to a biased distribution of attention across senses in favor of emotionally meaningful stimuli. For example, there is evidence that both aversive (e.g., Y. M. Wang et al., 2019) and pleasant (Y. Wang et al., 2022) sounds can bias spatial attention across modalities, leading to preferential allocation of visual attention to spatially congruent neutral stimuli. Furthermore, threat-related auditory cues

This article was published Online First November 13, 2025.
Yuan Chang Leong served as action editor.
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Materials (i.e., excluding stimuli that cannot be made available due to copyright restrictions), analysis code, and data are available at https://osf.io/mte32. The preregistration is available at https://aspredicted.org/LXG_2HM. All procedures performed in the present study were conducted in accordance with the Declaration of Helsinki, the guidelines of the German Research Foundation, and the guidelines of the Ethics Committee of the Faculty of Empirical Social Sciences of Saarland University. No specific formal approval for this study was required given that the study included voluntary participation, complete debriefing, and no harmful manipulation of any kind. Informed consent was obtained from all individual participants included in the study. All authors declare that they have no conflicts of interest. The research reported in this article was not funded by a third party (i.e., it was funded from the university budget).

Timea Folyi played a lead role in project administration, supervision, validation, and writing–original draft, a supporting role in data curation and resources, and an equal role in conceptualization, formal analysis, investigation, methodology, software, visualization, and writing–review and editing. Alexandra Alles played a lead role in data curation and resources, a supporting role in writing–original draft, and an equal role in formal analysis, investigation, methodology, software, visualization, and writing–review and editing. Dirk Wentura played a supporting role in project administration, supervision, and writing–original draft and an equal role in conceptualization, formal analysis, methodology, and writing–review and editing.

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have been suggested to broaden the distribution of spatial attention (Andrzejewski & Carlson, 2020). In addition, converging emotional information from different modalities, such as the concurrent presentation of unpleasant sounds and unpleasant images, may bias visual attention toward emotional images, as also suggested by the results of Gerdes et al. (2021).

Specifically, Gerdes et al. (2021) examined the influence of negative affective valence on the spontaneous orienting of visual attention. Therefore, they investigated the attentional bias toward unpleasant pictures in a free-viewing paradigm in which participants had no assigned task. Unpleasant and neutral pictures of complex scenes were presented in horizontally aligned pairs, while fixation behavior was measured by eye-tracking, a direct measure of the allocation of overt attention (e.g., Wright & Ward, 2008). On each trial, it was recorded which picture of the pair received preferential attention. The most important attentional process, the initial orientation, was operationalized via the first fixation, that is, whether it landed on the unpleasant or the neutral picture. The second attentional process of interest was sustained attention, measured as the duration of fixations on each picture of the pair excluding the initial fixation (i.e., dwell time; see Gerdes et al., 2021).[1]

Importantly, Gerdes et al. (2021) were not merely interested in testing whether attention is preferentially allocated to the unpleasant over the neutral picture of the pairs, but rather in the influence of several factors on this process. Each of these has previously attracted considerable theoretical interest. First, in line with the assumption of a leftward bias of emotional attention, the authors aimed to investigate the effect of presenting the emotional picture in the left versus the right hemifield. This research question is rooted in the right hemisphere hypothesis of emotion, which proposes that the right hemisphere would be specialized for the perception, attention to, expression, and experience of emotion (e.g., Ahern et al., 1991; Baijal & Srinivasan, 2011; for reviews, see Alves et al., 2008; Demaree et al., 2005; Gainotti, 2024). This assumption is also consistent with research suggesting a valence-specific difference in lateralized brain activity. Specifically, it has been suggested that there is a right versus left anterior specialization for processing and experience of negative versus positive emotions (or withdrawal vs. approach motivation; e.g., Davidson, 1992; Tomarken et al., 1992; for recent findings, see, e.g., Mansueto et al., 2025; Monni et al., 2022; for a review, see Harmon-Jones & Gable, 2018). However, recent reinterpretations of this hypothesis caution against oversimplification and emphasize its context dependence (e.g., Stanković, 2021). In line with these considerations, network-based models propose that emotion processing relies on multiple interrelated neural systems that may each show lateralized activity, but not necessarily the same pattern of lateralization (Palomero-Gallagher & Amunts, 2022). Thus, in line with the more general right hemisphere hypothesis of emotion processing, Gerdes et al. hypothesized that an attentional bias toward unpleasant pictures would be enhanced when they are presented on the left. Notably, this research question aimed to replicate the main finding of the study by Alpers (2008), using a comparable methodology.

Second, the main research question of Gerdes et al.'s (2021) study was aimed at possible cross-modal influences on the attentional bias to unpleasant pictures, namely, whether spatial and affective congruence with an accompanying sound would potentiate the allocation of visual attention to the unpleasant picture. To investigate this question, they presented concurrent unpleasant or neutral sounds on the left or right side. While there is ample evidence for interactions between visual and auditory attention based on spatial correspondence (for a review, see Driver & Spence, 1998), the role of emotional salience and emotional congruence in this process is an intriguing question, one that follows plausibly from the current literature on auditory and cross-modal information processing. Sounds are perceived near omnidirectionally, from a great distance, and their salient information is often reported to attract attention when selective attention is initially directed elsewhere (e.g., Dalton & Spence, 2007; Murphy et al., 2013). Sounds may therefore be highly effective in conveying critical emotional information when the visual environment is complex and visual attention is initially focused elsewhere. While accumulating evidence indicates that sounds can indeed guide visual attention to semantically congruent visual information (e.g., when referring to the same semantic object; Folyi et al., 2025; see also Iordanescu et al., 2010), evidence for the role of convergent auditory emotional information in guiding visual attention remains limited. Furthermore, the limited evidence points to an attentional facilitation that may be confounded by semantic congruence or, even more specifically, by a congruence between visual stimuli and naturally co-occurring characteristic sounds. Examples include presenting facial expressions and emotional vocalizations (e.g., Paulmann et al., 2012; Rigoulot & Pell, 2012) or semantically related complex images and sounds (Hu et al., 2024). An important novelty of the study by Gerdes et al. is that they targeted the possible influence of converging cross-modal emotional information on the guidance of visual attention by presenting complex scene pictures and auditory scenes without (unequivocal) semantic congruence.

Thus, Gerdes et al. (2021) investigated the effect of the presentation location of the pictures (in light of the right hemisphere hypothesis described above) and the presentation location and valence (i.e., unpleasant vs. neutral) of the sounds. Their novel question was whether spatial and valence (in)congruency with the unpleasant picture would affect the attentional bias toward these pictures. To test these effects, they analyzed the number of first fixations as a function of the fixed valence of the pictures (i.e., whether they were unpleasant or neutral) and several variable features of the picture presentation (position of the unpleasant picture: left vs. right; position of an accompanying sound: same vs. opposite to the unpleasant picture; valence of the sound: unpleasant vs. neutral). The authors analyzed the data as follows: First, for a given cell of the design of the variable factors, they counted the number of first fixations on the unpleasant picture and the number of first fixations on the neutral picture. Next, the sum of first fixations on the neutral picture was subtracted from the sum of first fixations on the unpleasant picture, and the resulting difference was divided by the total number of trials in that cell. This gives an index between +1 (i.e., *the unpleasant picture always receives the first fixation*) and −1 (i.e., *the neutral picture always receives the first fixation*). The authors called this dependent variable the bias score (BS). Their first result, that the average BS across all variable conditions is significantly positive ($M = 0.06$),

[1] As the main interest of the study was in the distribution of first fixations, we focus on this measure in the following. However, our argumentation below is also applicable to the measurement of dwell time. Of note, in the study of Gerdes et al. (2021), the results of dwell time largely mirrored the results of first fixations discussed below.

can be clearly interpreted in terms of valence. Thus, there is a significant tendency to fixate on the unpleasant picture first, compared to the neutral picture. This is consistent with previous findings showing that negative stimuli can attract preferential attention in the general population (for a review, see Yiend, 2010).

The study by Gerdes et al. (2021) focuses on the moderation of the BS by the variable factors. In this regard, three significant effects emerged in analyses with BS as the dependent variable. The authors found that (a) picture position had a main effect: A bias toward unpleasant pictures was more pronounced when unpleasant pictures were presented in the left ($M = 0.34$) than when they were presented in the right visual hemifield ($M = -0.22$). (b) BS was larger when a sound (regardless of valence) and the unpleasant picture appeared on the same side ($M = 0.18$) than on opposite sides ($M = -0.05$). (c) Finally, they found an interaction between picture position and sound valence; that is, the left–right difference in the BS (see above) is slightly increased when the sound was unpleasant.

Gerdes et al. (2021) interpreted these findings with a focus on the unpleasantness of the pictures and their processing characteristics: (a) as a leftward bias for emotional pictures, corresponding to a right hemisphere advantage for emotional processing; (b) that an attentional bias to unpleasant pictures is enhanced by spatial congruence with (any) sounds; and (c) that there is a complex cross-modal interaction between picture and sound valence and picture position. Specifically, they interpreted this interaction as emotional congruence between sounds and pictures increasing the bias to unpleasant pictures when they appear on the left. Taken together, these claims suggest that attention is systematically biased toward one side of visual space in order to detect emotionally significant cues, leading to a relative neglect of equally salient emotional stimuli presented on the opposite side. They further imply that spatial congruence with any sound will amplify a visual emotional attention bias. Moreover, a systematic bias of emotional attention toward the left side of space would be further reinforced by cross-modal emotional cues. These are far-reaching claims within the field of emotion and attention, as they point to a considerable rigidity in the emotional attention system (i.e., a systemic leftward bias and its enhancement). In the present article, we argue that such far-reaching claims must be evaluated using fair and methodologically robust approaches.

Specifically, we argue that in order to interpret these results in terms of Gerdes et al.'s (2021) interpretation, it is necessary to introduce an appropriate baseline condition against which the effects of the variable features can be tested. However, when using an incomplete design with only unpleasant–neutral pairs, any interpretation of the results (a) to (c) can be made without reference to the valence of the pictures. In the following section, we will develop this argument and then propose a solution, that is, a proposal for the design of the present study.

Let us assume a design analogous to that of Gerdes et al. (2021), but presenting pairs of neutral pictures. Thus, contrasting picture valence is no longer included as a fixed feature. With negative pictures no longer included, the study would instead examine how the first fixations are influenced by the variable features of stimulus presentation, such as picture position, sound location, and sound valence. A straightforward strategy to analyze the data would be to represent the dependent variable with two rows in the data, one per picture of the pair. This would indicate for each trial and each picture whether it received the first fixation or not. But since only one picture per trial can be fixated on first, one of the two rows is redundant. Therefore, we could randomly divide the full picture set into two subsets: Set A and Set B. Each trial then presents one A- and one B-picture in pairs. Given this strategy, each trial has only one entry, for example, referring to the A-picture. For each condition, we can then simply count the number of trials in which the A-picture received a first fixation and the number of trials in which the B-picture received a first fixation. We then subtract the latter from the former and divide this difference by the total number of trials for that condition. We obtain a BS′, a BS ranging from +1 (i.e., *the A-set picture always receives the first fixation*) to −1 (i.e., *the B-set picture always receives the first fixation*). Furthermore, any potential accidental differences between the A and B sets can be eliminated by using a counterbalancing design in which two arbitrary sets of neutral pictures alternately serve as the A and B sets. As a result, the average BS′ score will necessarily be zero, unlike a positive BS, which indicated the basic bias for unpleasant pictures (see above). Nevertheless, BS′ can be analyzed with regard to the position of the A-picture, whether a sound appears on the same side as the A-picture, and similar factors of the stimulus presentation.

Note that differences in the BS′ across conditions cannot be interpreted with reference to the content of the A- and B-pictures: Both sets are neutral, and counterbalancing ensures that there are no differences between them. For example, if the average BS′ is larger for the left location than for the right location, this simply means that the first fixation more often lands on the left picture, regardless of its content. Considering the effect of sound location, if the BS′ is higher when the sound is presented on the same side as the reference picture, this can be interpreted as a location compatibility effect. Finally, if the left–right difference in BS′ is larger when an unpleasant (vs. neutral) sound accompanies the picture, this can be interpreted as the unpleasant sound biasing the initial gaze toward the left picture over the right.

Of course, as Gerdes et al. (2021) were interested in biased attention toward unpleasant pictures, they used a set of unpleasant pictures and a set of neutral pictures. Accordingly, instead of our BS′, they calculated a BS. In our example, the mean value of BS′ is necessarily zero because it is based on splitting a homogeneous set of neutral pictures and counterbalancing their assignment to A and B sets. By contrast, the BS used by Gerdes et al. is meaningful, since positive values directly indicate a bias toward unpleasant pictures. Thus, their reported average of BS = 0.06 reflects a small but genuine bias toward unpleasant pictures. Our interpretation here is consistent with that reported by Gerdes et al.

However, the interpretation of Gerdes et al. (2021) goes further. They interpret the moderation of BS by the variable factors as evidence for the moderation of an emotional attention bias. As argued above, while the mean BS can be interpreted as a genuine emotional attention bias, its moderation by the variable factors can be interpreted in the same way as the moderation effects on our BS′. Thus, there is no need to refer to the distinction between unpleasant and neutral pictures. Consequently, based on the study of Gerdes et al., it remains an open question whether picture valence itself plays a role in results involving variable factors.

Of course, it is *possible* that the variable factor results for BS are more pronounced due to picture valence than those for the BS′. The questions raised by Gerdes et al. (2021) about the influence of presentation location and spatial and affective cross-modal (in)congruence on an emotional attention bias are intriguing and theoretically relevant. However, we argue that they can only be answered by specifically targeting such comparisons. Thus, crucially, the design used by Gerdes et al. (see also Alpers, 2008) requires the addition of an essential factor, namely, the variation of whether an unpleasant–neutral picture pair or a neutral–neutral picture pair is presented. The present study aimed to demonstrate the necessity of such a baseline condition.

Taken together, our suggestion is to implement a complete design by adding a neutral–neutral condition to the design used by Gerdes et al. (2021), which is adapted in the present study. Our specific aim was to test whether the effects of the variable factors emerge specifically in the unpleasant–neutral condition or also occur when only neutral picture pairs are presented. Regarding the variable factors, there were three significant effects by Gerdes et al., two of which had an effect size allowing for a high-powered replication with a manageable sample size in the present study. However, the interaction between sound valence and picture position was comparably weak ($d_Z = 0.29$); we therefore refrained from including this effect in our study plan.[2]

The Present Study

The present study aimed to directly test whether the effects of the variable factors on the BS, as calculated by Gerdes et al. (2021; see also Alpers, 2008), emerge specifically in a setting where unpleasant pictures are paired with neutral pictures. Alternatively, if these results can also be found for valence-invariant, neutral–neutral picture pairs, we can conclude that the emergence of these effects does not depend on the valence combination of the picture pairs. Our specific aim was to test whether the effects on attentional BS could be replicated with neutral–neutral pairs. Furthermore, if this is the case, a potential numerical difference in magnitude between an effect for unpleasant–neutral pairs and that for neutral–neutral pairs can be used as an estimate for high-power follow-up studies that address this difference. As argued above, such an approach would put to the test whether the effect reflects a genuine influence on emotional attention processes.

To this end, we conducted an extended replication of the study by Gerdes et al. (2021) by adding pairs of neutral pictures in addition to the unpleasant–neutral pairs of the original study. In line with our preregistration, we focused on the number of first fixations of the eye-tracking data as our main eye-tracking measure, which indicates the initial allocation of visual attention. (For the sake of completeness, we additionally report all effects on the subsequent dwell time, which reflects sustained attention.) For the unpleasant–neutral pairs, we calculated the BS for the number of first fixations as the dependent variable, following Gerdes et al. (see also above). Thus, BS is an index ranging from +1 (i.e., *the unpleasant picture always receives the first fixation*) to −1 (i.e., *the neutral picture always receives the first fixation*). For the neutral–neutral pairs, we calculated an arbitrary BS′ as explained above. Specifically, first, we divided a set of neutral pictures into two arbitrary subsets, which we call X and Y sets of pictures (see Methods). We then counterbalanced across participants whether the arbitrary X and Y sets of pictures were assigned to the arbitrary denotations of A versus B sets when calculating the BS′ in favor of the A-set denotation. As explained above, for each condition of the variable factors, we subtracted the number of first fixations on B-set pictures from the number of first fixations on A-set pictures and divided this difference by the total number of (valid) trials for the condition. Thus, BS′ is an index ranging from +1 (*which means that the A-set picture always receives the first fixation*) to −1 (*which means that the B-set picture always receives the first fixation*).

The counterbalanced assignment of the X- and Y-picture sets to the A- versus B-set denotations was essential for testing the effects of the variable features on a BS that is *by design* not related to the content or other characteristics of the picture sets. Without this counterbalancing, BS′ in favor of the A-pictures could have been systematically influenced by incidental differences between the two neutral sets (e.g., perceptual features or semantic content) that are unrelated to our hypotheses. The counterbalancing ensures that the effects of the variable features of picture presentation on BS′ can be interpreted independently of such differences.[3] Needless to say, due to the counterbalancing of X and Y sets, the expected value for the *average* BS′ across conditions will be zero.

However, if we were to leave it at that, this new design would introduce a confound: The unpleasant pictures would now be a rare stimulus category compared to the neutral pictures. Therefore, in order to have a balanced design in terms of picture valence, another valence-invariant condition consisting of unpleasant–unpleasant picture pairs was added. These picture pairs served dominantly as filler trials. Thus, as preregistered, in the following, we focus on the valence–neutral and neutral–neutral picture pair conditions. The results of the unpleasant–unpleasant condition will be reported in the Appendix.

Overall, we expected a positive mean BS in line with the findings of Gerdes et al. (2021). We then tested whether the effects of the variable factors on BS would be replicated in the present study. Specifically, we expected that (a) BS would be more pronounced when unpleasant pictures were presented on the left compared to the right picture position and (b) BS would be more pronounced when the sound and the unpleasant picture appeared on the same side compared to the opposite side. Critically, we also tested Hypotheses 1 and 2 on the arbitrary BS′, with the A-set pictures being the reference set instead of the unpleasant pictures. Specifically, we tested if (a) BS′ would be more pronounced when the A-set pictures were presented on the left compared to the right and if (b) BS′ would be more pronounced when the sound and the A-set picture appeared on the same side compared to the opposite side. If the effects described in Hypotheses 1 and 2 also emerge for BS′, this will support our claim that the interpretation of BS in terms of valence is overstretched. As a secondary analysis, we also test the corresponding effects on the secondary eye-tracking measure, that is, dwell time.

Note that except for the mentioned changes, we followed as closely as possible the study by Gerdes et al. (2021).[4] Especially, we

[2] Note that a simple replication attempt would need at least $N = 157$ (in a single-session eye-tracking lab experiment) to achieve a power of $1 - \beta = .95$ ($\alpha = .05$).

[3] Note that as we considered this counterbalancing an inherent and self-suggesting step of good experimental practice, we did not explicitly mention it in our preregistration.

[4] There were two exceptions that did not affect the experimental part of the study. First, we did not assess participants' depressive mood, state and trait anxiety, and positive and negative affective states. Gerdes et al. (2021) did not analyze these variables in relation to the BS in their study. As our study involved a sample of students, we did not consider it necessary to screen our sample on these measures. Second, an independent sample of $N = 20$ rated the presented pictures and sounds for valence and arousal. This rating study also served as a basis for selecting additional stimuli with comparable valence and arousal ratings, as used by Gerdes et al. (see Material).

followed one procedural detail that did not play a role in our report until now. Although the main focus of the study was on the audio-visual trials, a block of pure visual trials was administered by Gerdes et al. as well. The order of blocks was counterbalanced. We also included a visual-only block and kept the feature of order balancing.

Method

Transparency and Openness

The experiment was preregistered at https://aspredicted.org/LXG_2HM. We report how we determined our sample size, all data exclusions, all manipulations, and all measures in the study. The data collection took place between November 2022 and May 2023. Materials (i.e., excluding the experimental stimuli that cannot be made available due to copyright restrictions), analysis code, and data are available at https://osf.io/mte32. Data were prepared for analysis using BeGaze (Version 3.5.74) and analyzed using SPSS (Version 29.0.2).

The study was conducted in accordance with the Declaration of Helsinki and the guidelines of the German Research Foundation and the Ethics Committee of the Faculty of Empirical Social Sciences of Saarland University. No specific formal approval was required for this study, as it involved voluntary participation, full debriefing about the study (i.e., including debriefing before the study about the fact that negative emotional pictures and sounds would be presented), and no harmful manipulation of any kind.

Participants

We based our power calculations on the effects found by Gerdes et al. (2021) on the main dependent variable, BS, for the number of first fixations (in the part of the study, i.e., in the audio-visual block). In Gerdes et al., the effect that congruence of sound location and location of the unpleasant picture enhanced the BS was $d_Z = 0.74$; the effect of the picture position of the unpleasant picture was $d_Z = 0.54$. To detect an effect of $d_Z = 0.54$ with power $1 - \beta = .95$ ($\alpha = .05$, two-tailed), a sample size of at least $N = 47$ is needed. To have a complete counterbalancing scheme, we decided on the final sample size of $N = 48$.

Participants were 48 students from Saarland University, recruited based on the following criteria: native or fluent German speaker, over 18 years of age, normal or corrected-to-normal vision with contact lenses, and self-reported normal hearing. Participants received €7.50 or course credits as compensation. Data from four additional participants could not be adequately recorded due to technical problems with the eye tracker and were replaced by new participants. Data from three participants were excluded from the data analyses based on the preregistered criterion; that is, more than 40% of a participant's trials had to be removed. These trials were defined as either having no fixation on either of the two presented pictures or as having far-out dwell time values according to the individual distribution of dwell times (Tukey, 1977).

The final sample of $N = 45$ participants consisted of 32 women and 13 men with a mean age of 22.4 years (SD = 4.9 years). Participants were predominantly right-handed ($n = 39$ right, $n = 6$ left), and the dominant eye was predominantly the right eye ($n = 33$ right, $n = 11$ left, and $n = 1$ showed ambiguous results and was tested using the right eye).[5] With $N = 45$, power was still $1 - \beta = .94$ for the smaller effect; for the larger effect, it was $1 - \beta = .998$. For the basic BS (i.e., the effect that the first fixation landed more often on the unpleasant picture; $d_Z = 0.52$), power was $1 - \beta = .93$.

Design

The audio-visual block followed a 3 (picture pair: unpleasant–neutral vs. neutral–neutral vs. unpleasant–unpleasant) × 2 (position of the unpleasant/A-set picture: left vs. right) × 2 (position of the sound: same vs. opposite to the unpleasant/A-set picture) × 2 (sound valence: unpleasant vs. neutral) repeated measures design. The visual-only block followed a 3 (picture pair: unpleasant–neutral vs. neutral–neutral vs. unpleasant–unpleasant) × 2 (position of the unpleasant/A-set picture: left vs. right) repeated measures design. The presentation order of the visual-only and audio-visual task blocks was counterbalanced between participants. Furthermore, in the neutral–neutral and unpleasant–unpleasant picture pair conditions, the assignment of the arbitrary X versus Y sets of pictures (see Material) to the arbitrary A- versus B-set labels was counterbalanced between participants. In addition, for the audio-visual block, three sound sets were used (see Material), and the assignment of the sound sets to the picture pair conditions was counterbalanced between participants. Overall, our hypotheses relate to the unpleasant–neutral and neutral–neutral picture pair conditions, while trials with the unpleasant–unpleasant picture pairs were mainly introduced as fillers to avoid valenced pictures being a rare category compared to neutral pictures.

Material

Apparatus

The eye movements of the participants' dominant eye were recorded with an SMI Hi-Speed Eye-Tracker with a sample rate of 1,250 Hz and a spatial resolution of 0.01° (manufacturer information). We used the recording software iView X (Version 2.8.43) with a calibration area of 1,920 × 1,080 px. The calibration accuracy was set to a maximum of 1° visual angle deviation on the horizontal or vertical coordinate. On average, the calibration accuracy was M = 0.38° visual angle (SD = 0.25°).[6] The parameters for fixation detection were a maximum dispersion value of 2.02° visual angle (approximately 57 pixels) and a minimum fixation duration of 80 ms, consistent with Gerdes et al. (2021). The pictures were displayed on a monitor with a size of 27 in. and a resolution of 1,920 × 1,080 pixels at a refresh rate of 120 Hz. Viewing distance was 50 cm. Sounds were presented through loudspeakers positioned right and left of the monitor at the same distance from the participant as the monitor. Sounds were presented at a sound pressure level of approximately 72 dB sound pressure level. The experimental software PsychoPy (Version 1.90.1; Peirce et al., 2019) was used for stimulus presentation.

[5] Other demographic measures (e.g., ethnicity or socioeconomic status) were not considered significant for the main research question and were therefore not collected. This is also addressed in the Constraints on Generality of the Findings section.

[6] In a recalibration during the experiment, the calibration accuracy of one reference point by one participant was accepted with a deviation of 1.22°. All other calibration points showed a maximum deviation of less than 1.00°.

Stimuli

A total of 144 pictures from the International Affective Picture System (Lang et al., 2008), the Open Affective Standardized Image Set (Kurdi et al., 2017), and the Nencki Affective Picture System (Marchewka et al., 2014) were used in the present study. Identification codes of the pictures are reported in Supplemental Table S1. In order to achieve an exact replication of Gerdes et al. (2021), we used the same 48 pictures (i.e., a set of 24 unpleasant and a set of 24 neutral pictures) for the unpleasant–neutral condition. In addition, for each of the unpleasant–unpleasant and neutral–neutral picture pair conditions, two further sets of pictures were selected: For the unpleasant–unpleasant condition, each set consisted of 24 unpleasant pictures, and for the neutral–neutral condition, each set consisted of 24 neutral pictures (labeled as X and Y sets in each condition). Consistent with Gerdes et al., all pictures were in landscape format and belonged to three different semantic categories: depicting either humans, animals, or inanimate objects (eight pictures per category in each set). Some of the International Affective Picture System pictures had a black frame around the picture. These pictures were cropped to remove the frame while maintaining their length-to-width ratio. The selected pictures were matched in valence and arousal (for mean normative valence and arousal ratings, see Table 1) and, furthermore, for brightness and contrast to the unpleasant and neutral pictures used in Gerdes et al. (for mean brightness and contrast values, see Supplemental Table S2).

In addition to the normative valence and arousal ratings, valence and arousal ratings were obtained for a preselection of pictures and sounds from an online study. This was conducted prior to the eye-tracking study with $N = 20$ additional participants (10 female; aged 18–35 years; all German native speakers). Valence (from 1 = *very unpleasant* to 9 = *very pleasant*) and arousal ratings (from 1 = *completely unaroused* to 9 = *completely aroused*) were obtained using the Self-Assessment Manikin (Bradley & Lang, 1994). The study was programmed in PsychoPy3 (Version 2021.1.4, https://www.psychopy.org) and hosted via the online study platform Pavlovia (https://pavlovia.org/). Recruitment was carried out via Prolific (https://www.prolific.com), and participants were paid £6 (approximately €7) for a duration of 40 min. The preselected sets of pictures and sounds overlapped strongly with the pictures and sounds used in the eye-tracking study. Two pictures (i.e., one neutral and one negative picture selected for the present study) were replaced for the final stimulus selection of the main study based on these rating data. These two pictures showed overlapping ratings with the other valence category and were replaced by new pictures based on their normative ratings. There was a high positive correlation between the ratings obtained in this online study and the normative ratings of the pictures: for valence: $r(144) = .959$, $p < .001$; for arousal: $r(144) = .898$, $p < .001$. For the sake of completeness, mean valence and arousal ratings from this study are presented in the Supplemental Material (for the pictures, see Supplemental Table S4). In the eye-tracking study, the pictures were presented in pairs at a visual angle of 17.04° × 14.82° (horizontal × vertical) at a distance of 12.42° from each other, at a distance of approximately 50 cm from the screen. These data corresponded approximately to the description in Gerdes et al. (2021).

The audio-visual block presented 72 sounds selected from the Mannheim Affective Sound Inventory (Gerdes, 2020) and the International Affective Digitized Sounds (Bradley & Lang, 2007).[7] Identification codes of the sounds are reported in Supplemental Table S3. One set of 12 unpleasant and 12 neutral sounds were the sounds used by Gerdes et al. (2021); in addition, we selected two additional sets, each consisting of 12 unpleasant and 12 neutral sounds. These two sets were selected to match in valence and arousal the set used by Gerdes et al. (for mean normative valence and arousal ratings, see Table 2). As for the pictures, the sounds were also chosen from three categories: human, animal, or inanimate sounds (in each category, there were four sounds per set and per valence). Segments with a length of 2,000 ms were extracted from the International Affective Digitized Sounds, with the aim of preserving the semantic content of the sound, in line with Gerdes et al. Similarly as with the pictures, three of the sounds from the additionally selected negative sounds and six of the additionally selected neutral sounds were replaced for the final stimulus selection based on the ratings from our online study. These sounds showed overlapping valence and/or arousal ratings with the other valence categories. The new sounds were selected based on their normative ratings. The correlations of the sound ratings in this online study with the normative ratings were again high, both for valence, $r(72) = .877$, $p < .001$, and for arousal, $r(72) = .754$, $p < .001$. Mean valence and arousal ratings of the sounds are presented in Supplemental Table S5. In the eye-tracking study, sounds were presented in mono audio format through speakers on either the left or the right of the screen (see Procedure).

Table 1
Mean Normative Valence and Arousal Ratings of the Pictures

		Rating *M* (*SD*)	
Valence category	Picture set	Valence	Arousal
Neutral	Gerdes et al. (2021)	5.32 (0.66)	3.32 (0.62)
	X set	5.48 (0.66)	3.51 (0.52)
	Y set	5.49 (0.63)	3.37 (0.57)
Unpleasant	Gerdes et al. (2021)	2.45 (0.55)	6.03 (0.45)
	X set	2.65 (0.52)	5.95 (0.79)
	Y set	2.52 (0.48)	6.01 (0.52)

Note. The table shows the mean normative valence (1 = *very negative/very unpleasant* to 9 = *very positive/very pleasant*) and arousal ratings (1 = *completely unaroused/very low* to 9 = *completely aroused/very high*) of the pictures (for International Affective Picture System, Lang et al., 2008; for OASIS, Kurdi et al., 2017; for Nencki Affective Picture System, Marchewka et al., 2014) for each valence category and picture set. There were 24 images per set and valence category. Gerdes et al. (2021) refers to the image set used in Gerdes et al. and for our unpleasant–neutral replication condition; the neutral X/Y sets were used for our neutral–neutral condition with X set = A set (Y set = B set) for half the participants and the reversed assignment for the other half; in the same manner, the unpleasant X/Y sets were used for our unpleasant–unpleasant condition (which mainly served as filler trials; see the Appendix for results related to this condition). The normative valence and arousal ratings of the Open Affective Standardized Image Set (OASIS) database (Kurdi et al., 2017) used a scale of 1 = *very negative/very low arousal* to 7 = *very positive/very high arousal*. For better comparability, the OASIS rating data were recoded to a scale of 1–9.

[7] Two 2,000-ms excerpts were extracted from two sounds selected from the International Affective Digitized Sounds database. The extracted segments were clearly distinguishable.

Table 2
Mean Normative Valence and Arousal Ratings of the Sounds

Valence category	Sound set	Rating *M* (*SD*)	
		Valence	Arousal
Neutral	Gerdes et al. (2021)	5.29 (0.39)	3.90 (0.71)
	b set	5.18 (0.33)	4.29 (0.63)
	c set	5.37 (0.50)	4.29 (0.41)
Unpleasant	Gerdes et al. (2021)	2.76 (0.47)	6.47 (0.51)
	b set	2.89 (0.51)	6.53 (0.43)
	c set	2.80 (0.41)	6.49 (0.39)

Note. The table shows the mean normative valence (1 = *very negative/very unpleasant* to 9 = *very positive/very pleasant*) and arousal ratings (1 = *completely unaroused* to 9 = *completely aroused*) of the sounds (for International Affective Digitized Sounds, Bradley & Lang, 2007; for Mannheim Affective Sound Inventory, Gerdes, 2020) for each valence category and sound set. There were 12 sounds per set and valence category. Gerdes et al. (2021) refers to the sound set used in Gerdes et al.; b/c set = selection of additionally created sets.

Procedure

Each participant gave informed consent prior to the experiment. After completing a demographic questionnaire and determining which eye was the dominant eye, participants were seated at the eye tracker in front of the experimental monitor. The height of the chin rest, chair, and table, as well as the eye-tracking camera, was adjusted to center the dominant eye. The iView recording software was used to focus the image. Thereafter, a standard 13-point calibration of the eye tracker was administered. Participants were instructed to avoid moving their head and to keep their eyes on the screen for the duration of the experiment. Participants were not given an explicit task for the experiment, in line with the procedure of Gerdes et al. (2021) and the usual procedure in the free-viewing paradigm (Armstrong & Olatunji, 2012; Waechter et al., 2014).

Following the procedure of Gerdes et al. (2021), the experiment consisted of an audio-visual block and a visual-only block, and the order of presentation of these blocks was counterbalanced between participants. The audio-visual block was divided into three experimental blocks of 72 trials each, each lasting approximately 5 min. Participants could take a short break between blocks. In a block of 72 trials, each possible combination of the picture pair condition, picture position (i.e., position of the unpleasant/A-set picture), sound position, and sound valence was presented three times, randomly intermixed. In the visual-only part of the study, each possible combination of the picture pair and picture position (i.e., position of the unpleasant/A-set picture) condition was repeated 12 times, randomly intermixed, resulting in 72 visual-only trials presented in a single block. Three practice runs were conducted before the audio-visual and the visual-only blocks, respectively. A total of 288 experimental trials (audio-visual: 216 trials; visual-only: 72 trials) were thus administered, with each picture being presented three times in the audio-visual and once in the visual-only block. The entire session lasted approximately 30–45 min.

The procedure of a trial was identical to that in Gerdes et al. (2021) and is depicted in Figure 1 (i.e., for the audio-visual block). Each trial began with the presentation of a fixation cross in the center of the screen for 1,000 ms, followed by the stimulus presentation for 2,000 ms. The stimulus presentation consisted of the presentation of a horizontally aligned picture pair (i.e., one unpleasant and one neutral picture, two neutral pictures, or two unpleasant pictures, according to the picture pair condition). The pictures in the pairs were of the same semantic category (i.e., animal, human, inanimate object). Whereas in the visual-only block only pictures were presented, in the audio-visual block, a neutral or unpleasant sound was additionally presented from either the left or the right side of the screen via a loudspeaker during the presentation of the visual stimuli. The sound had the same category as the picture pair (i.e., animal, human, or inanimate object). The stimulus presentation was followed by an intertrial interval of 800 ms.

Data Preparation

Fixations, saccades, and blinks of the eye-tracking data were detected by using BeGaze (Version 3.5.74). Additionally, relevant areas of interest (AOIs) were defined for the left and the right picture locations. These AOIs were defined as 10% larger than the actual representation of the images (horizontal × vertical: 18.74° × 16.30° visual angle), following Gerdes et al. (2021). The AOI for the fixation cross was defined between these areas.

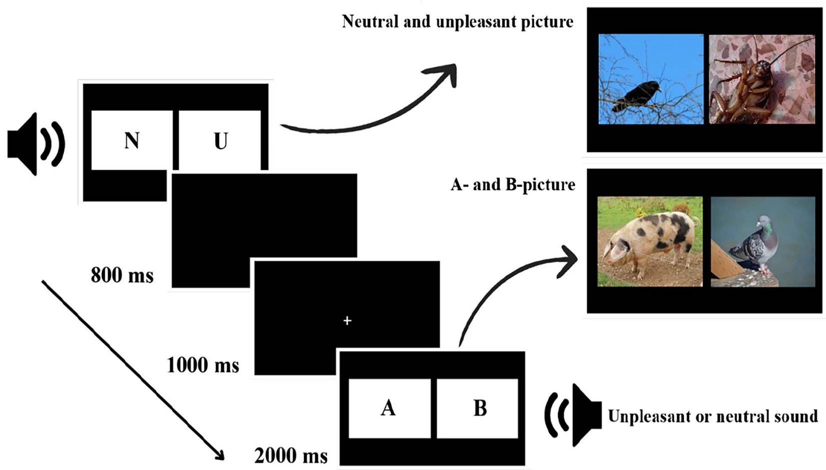
The main dependent variable was the number of the first fixations. A first fixation was defined as the first fixation on one of the AOIs for the pictures following the first saccade away from the fixation cross (thus, trials in which the first fixation on one of the pictures occurred after a blink were excluded from the analyses of first fixations; see also Alpers, 2008). Dwell time was defined for each of the pictures in the pairs as the total duration of the fixations, excluding the duration of the first fixation, on the respective AOI. As preregistered, far-out dwell time values according to the individual distribution of dwell times (i.e., three interquartile ranges below the first or above the third quartile; Tukey, 1977) were excluded.

For both dependent variables, a BS for the unpleasant–neutral picture pairs and a BS (BS′) for the neutral–neutral (and unpleasant–unpleasant) picture pairs were calculated following the method of Gerdes et al. (2021). Accordingly, for each participant and condition, the BS and BS′ of the first fixations were calculated as the sum of the initial fixations that landed on the unpleasant picture (for BS) or the A-set picture (for BS′) minus the sum of the initial fixations that landed on the neutral picture (for BS) or the B-set picture (for BS′), divided by the sum of the two (i.e., the total number of trials in the respective condition after excluding the invalid trials). For the dwell time, BS and BS′ were calculated for each trial as the total duration of the fixations on the unpleasant picture (for BS) or the A-set picture (for BS′) minus the total duration of the fixations on the neutral picture (for BS) or the B-set picture (for BS′), divided by the sum of the two. BS and BS′ of the dwell times were then averaged for each condition for each participant.

Statistical Analyses

As preregistered, our main analysis addresses the unpleasant–neutral and neutral–neutral picture pair conditions, respectively. (The results of the corresponding analyses for the unpleasant–unpleasant condition are reported in the Appendix. In a nutshell, as expected, the two valence-invariant picture pair conditions showed comparable results.) Thus, we tested our main hypotheses on the BS and BS′ for the number of first fixations in the unpleasant–neutral and neutral–neutral picture pair conditions of the audio-visual block. In the unpleasant–neutral condition, we conducted a 2 (position of the unpleasant picture: left vs. right) × 2 (position of the sound: same vs. opposite to the unpleasant picture) × 2 (sound valence: unpleasant

Figure 1
Trial Sequence in the Audio-Visual Block



Note. The example depicts trials with unpleasant–neutral picture pairs (i.e., with one unpleasant and one neutral picture denoted by "U" and "N," respectively) and neutral–neutral picture pairs (i.e., with one picture presented from the arbitrary A set and one from the arbitrary B set of neutral pictures). The illustration features example pictures from the animal category taken from the Open Affective Standardized Image Set database (Kurdi et al., 2017). A neutral or unpleasant sound was presented synchronously with the pair of pictures through loudspeakers on the left or on the right side of the screen (i.e., on the same or opposite side of the unpleasant or the A-set picture). The trial sequence in the visual-only block was identical except that no sound was presented. Due to copyright restrictions, we only depict pictures from the Open Affective Standardized Image Set database (Kurdi et al., 2017). The pictures shown as examples for unpleasant and neutral pictures, Crow 2 (for neutral) and Cockroach 1 (for unpleasant), were used in the present study (see the Supplemental Material). The pictures shown as examples for A- and B-pictures, Pig 1 and Pigeon 5, were not used in the present study. These pictures have comparable normative valence and arousal ratings as the stimuli used in the study. See the online article for the color version of this figure.

vs. neutral) repeated measures analysis of variance (ANOVA) on the BS. The constant test of this analysis corresponds to the test of whether the BS averaged across all conditions of the variable factors is significantly different from zero (i.e., it tests for the replication of a general BS). In the neutral–neutral condition, we conducted the corresponding 2 (position of the A-set picture: left vs. right) × 2 (position of the sound: same vs. opposite to the A-set picture) × 2 (sound valence: unpleasant vs. neutral) repeated measures ANOVA on the BS′. (We also report the constant test of this analysis, which tests whether the mean BS′ is different from zero. Note that if it is significant, this would be an a priori false positive due to our design.)

In the visual-only block, we performed two corresponding analyses on the BS and BS′ in the unpleasant–neutral and neutral–neutral conditions, respectively. In the unpleasant–neutral condition, we conducted a one-way ANOVA with the position of the unpleasant picture (left vs. right) as a repeated measures factor on the BS of first fixations. An analogous one-way ANOVA was conducted on the BS′ of the neutral–neutral condition, with the position of the A-set picture (left vs. right) as a repeated measures factor. The constant tests of these analyses correspond to the tests of whether the average BS and BS′ are significantly different from zero in the visual-only block.

Furthermore, we performed the analogous analyses on the secondary eye-tracking measure, the subsequent dwell time (i.e., after the first fixations). BS/BS′ of the dwell time reflect the corresponding BS for sustained attention. Although it was not the aim of the present study to test differences between the picture pair conditions, an exploratory analysis comparing the unpleasant–neutral and neutral–neutral picture pair conditions of the audio-visual block is also reported for the interested reader.

Additionally, we added Bayes factors (BFs) to the main analyses, thus specifically to those analyses testing our central hypotheses regarding the BS and BS′ on the number of initial fixations in the unpleasant–neutral and neutral–neutral picture pair conditions of the audio-visual block. For each analysis, the null model (H_0) states that there is no difference between the respective conditions in the predicted direction. The alternative model (H_1) for the main analyses corresponds to the directional predictions specified in Hypotheses 1 and 2, respectively, regarding the BS and the BS′ of the neutral–neutral condition. For all reported *F* tests, there is one numerator degree of freedom. Using equivalence with a *t* test, BFs are reported from the corresponding Bayesian *t* tests.

In line with recent recommendations (e.g., Dienes, 2021), we employed informative priors based on predicted effect sizes. We used the effect sizes reported by Gerdes et al. (2021) as a reference for the main analyses on the BS and the equivalent analyses on the BS′. In line with the aforementioned recommendations and our directional predictions, we modeled the predictions of H_1 as a half-normal distribution with an *SD* equivalent to the expected size of the effect (i.e., d_Z). This model represents the more likely occurrence of smaller effects than larger effects. Using the notation $BF_{HN(0,\,x)}$, we refer to a BF in which the predictions of H_1 were modeled as a half-normal distribution with an *SD* of *x*, where *x* scales the expected effect size (Dienes, 2014).

Additionally, to inform future research, we present the BFs for the exploratory analyses that test for the differences between the unpleasant–neutral and the neutral–neutral conditions. The H_1 states that the differences described in Hypotheses 1 and 2 are more pronounced in the unpleasant–neutral condition than in the neutral–neutral condition. In contrast, the H_0 states that there is no difference between the respective conditions in the predicted direction. For these analyses, since we have no prior knowledge of the expected size of the effect, we used the default prior (i.e., Cauchy scale with scale parameter of $r = 1$).

We interpret BFs >3, >10, >30, and >100 as indicating moderate, strong, very strong, and extreme evidence for H_1, respectively; by symmetry, we interpret BFs <1/3, <1/10, <1/30, and <1/100 as indicating the corresponding level of evidence for H_0 (e.g., Schönbrodt & Wagenmakers, 2018; Schönbrodt et al., 2017). Accordingly, BFs between 1/3 and 3 are interpreted as inconclusive evidence. To indicate the robustness of these conclusions, we report robustness regions (RR) for each BF, indicating the range of expected effect sizes for the H_1 that would lead to the same conclusion as the one reported (i.e., evidence for H_1 of H_0 with a respective strength based on convention or inconclusive evidence; e.g., Lakens et al., 2020; we used R Version 4.5.0 to calculate the BF and RR, using the package *bfrr*, Version 0.0.0.9000, DeBruine & Dienes, 2022; and the package *BayesFactor*, Version 0.9.12-4.7, DeBruine & Dienes, 2022).

Results

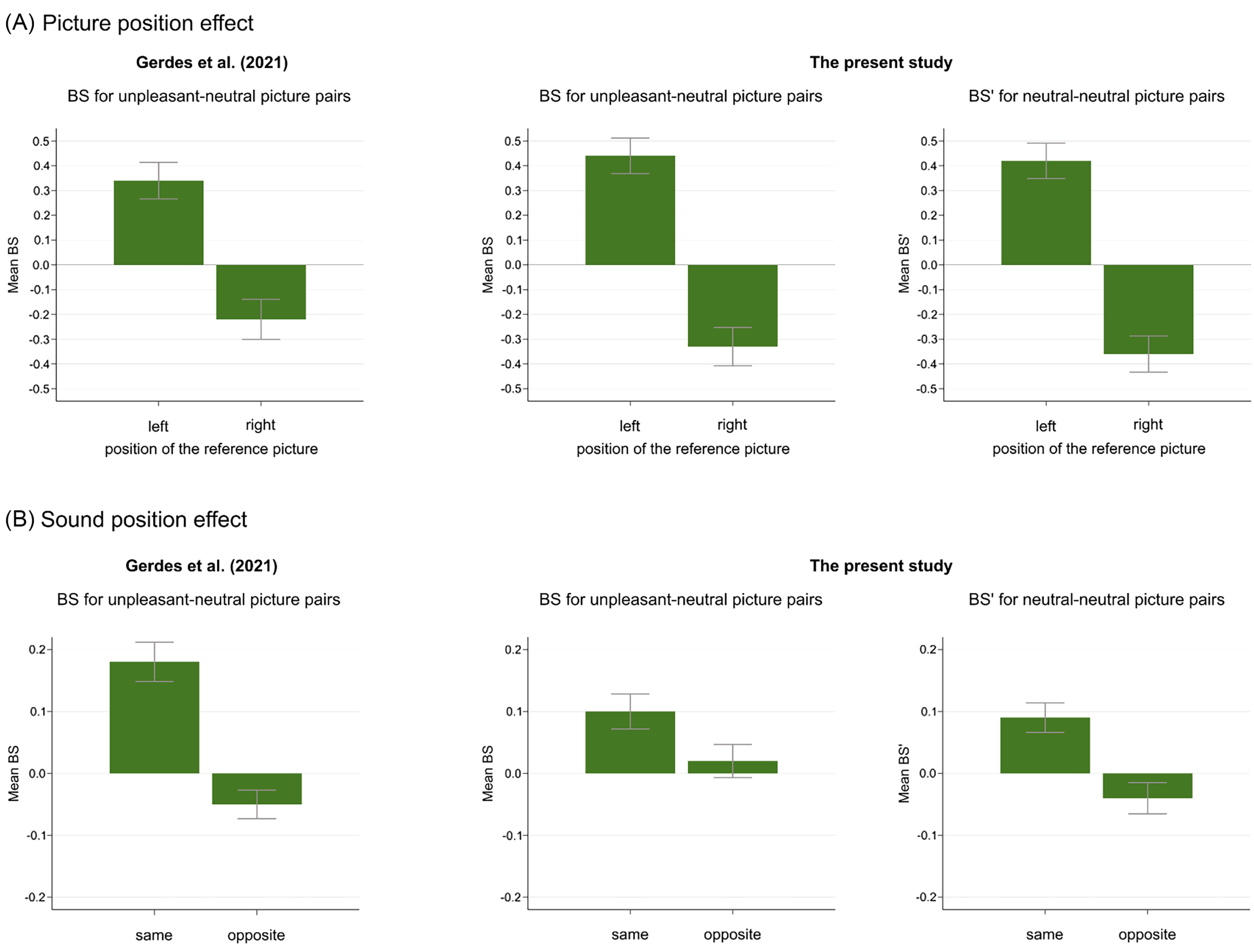
All effects referred to as statistically significant throughout the text are associated with p values below .05, two-tailed.

Number of First Fixations

Mean BS of the first fixations in the audio-visual block for each picture and sound position condition are presented in Figure 2, separately for the unpleasant–neutral (BS) and neutral–neutral

Figure 2

Mean BS of the First Fixations in the Audio-Visual Block in the Study of Gerdes et al. (2021; BS, Right Column) and in the Unpleasant–Neutral (BS, Middle Column) and Neutral–Neutral Picture Pair Conditions (BS′, Left Column) of the Present Study



Note. (A) Mean BS for the left versus right position of the reference picture (i.e., unpleasant picture for BS, A-set picture for BS′). (B) Mean BS for the same versus opposite sound position relative to the reference picture. Error bars represent the standard error of the means. BS = bias scores. See the online article for the color version of this figure.

picture pair conditions (BS′) of the present study. For comparison, we additionally depict the corresponding means for BS from the study of Gerdes et al. (2021). Figure A1 in the Appendix additionally depicts the individual data points from the present study.

Unpleasant–Neutral Picture Pairs

In the audio-visual block of the unpleasant–neutral condition, the 2 (position of the unpleasant picture: left vs. right) × 2 (position of the sound: same vs. opposite to the unpleasant picture) × 2 (sound valence: unpleasant vs. neutral) repeated measures ANOVA on the BS of the first fixations as the dependent variable revealed a significant constant effect, $F(1, 44) = 8.75$, $p = .005$, $\eta_p^2 = .166$ ($d_Z = 0.44$); $BF_{HN(0,\ 0.52)} = 23.40$, $RR_{B>10}$ [0.16, 1.70], indicating strong evidence for H_1. Thus, the BS averaged across all conditions ($M = 0.06$, $SD = 0.13$) is significantly above zero, replicating a general attentional bias toward unpleasant pictures. Corroborating the results of Gerdes et al. (2021), the analysis revealed a significant main effect of the picture position, $F(1, 44) = 28.81$, $p < .001$, $\eta_p^2 = .396$ ($d_Z = 0.80$); $BF_{HN(0,\ 0.54)} = 16{,}129$, $RR_{B>100}$ [0.15, >2], reflecting extreme evidence in favor of the H_1. As expected, this result corresponds to an increased BS for the left ($M = 0.44$, $SD = 0.48$) compared to the right position of the unpleasant picture ($M = -0.33$, $SD = 0.52$). Furthermore, the main effect of sound position was also significant, $F(1, 44) = 4.95$, $p = .031$, $\eta_p^2 = .101$ ($d_Z = 0.33$); $BF_{HN(0,\ 0.74)} = 3.89$, $RR_{B>3}$ [0.11, 1.01], showing moderate evidence for H_1. As expected, BS was larger when the location of the sound and the unpleasant picture were congruent ($M = 0.10$, $SD = 0.19$) than when they were presented in opposite locations ($M = 0.02$, $SD = 0.18$). No other significant main effect or interaction emerged, $F(1, 44) = 2.88$, $p = .097$, $\eta_p^2 = .061$, for the interaction of sound position and sound valence (all other Fs < 1, ps > .359). Thus, as can also be seen in Figure 2, we were able to replicate the essential findings of Gerdes et al. in the unpleasant–neutral condition.[8]

In the visual-only block, we conducted a one-way ANOVA on the BS of the first fixations with the position of the unpleasant picture (left vs. right) as a repeated measures factor. The constant effect, testing for a mean bias for unpleasant pictures, did not reach significance in this task block, which had a much lower number of trials than the audio-visual block, $F(1, 44) = 0.79$, $p = .378$, $\eta_p^2 = .018$ ($d_Z = 0.13$; $M = 0.03$, $SD = 0.21$, for the BS). Consistent with the results of the audio-visual block, the BS differed significantly between picture positions, $F(1, 44) = 14.65$, $p < .001$, $\eta_p^2 = .250$ ($d_Z = 0.57$), with higher BS when the reference picture was presented on the left side ($M = 0.35$, $SD = 0.60$) compared to the right side ($M = -0.30$, $SD = 0.60$).

Neutral–Neutral Picture Pairs

In the valence-invariant neutral–neutral condition, we performed the analogous analyses as in the unpleasant–neutral condition. Thus, in the audio-visual block, a 2 (position of the A-set picture: left vs. right) × 2 (position of the sound: same vs. opposite to the A-set picture) × 2 (sound valence: unpleasant vs. neutral) repeated measures ANOVA was conducted on the BS′ of the first fixations. As expected, this analysis did not yield a significant constant effect, $F(1, 44) = 3.01$, $p = .090$, $\eta_p^2 = .064$ ($d_Z = 0.26$; the average BS′ was $M = 0.03$, $SD = 0.11$). Mirroring the results with unpleasant–neutral picture pairs, a significant main effect of picture position emerged, $F(1, 44) = 30.36$, $p < .001$, $\eta_p^2 = .408$ ($d_Z = 0.82$); $BF_{HN(0,\ 0.54)} = 24{,}572$, $RR_{B>100}$ [0.15, >2], showing extreme evidence for H_1. As for the BS of the unpleasant–neutral condition, the BS′ for an arbitrary set was also significantly increased when the reference picture was presented on the left side ($M = 0.42$, $SD = 0.48$) compared to the right side ($M = -0.36$, $SD = 0.49$; see Figure 2). Furthermore, as for the BS with unpleasant–neutral pairs, the main effect of sound position was also significant on the BS′, $F(1, 44) = 11.55$, $p = .001$, $\eta_p^2 = .208$ ($d_Z = 0.51$); $BF_{HN(0,\ 0.74)} = 60.04$, $RR_{B>30}$ [0.21, 1.81], thus strong evidence for H_1. Again, BS′ was higher when the reference picture and the sound were spatially congruent ($M = 0.09$, $SD = 0.16$) than when they were incongruent ($M = -0.04$, $SD = 0.17$; see also Figure 2). No other significant main effect or interaction emerged, $F(1, 44) = 2.77$, $p = .103$, $\eta_p^2 = .059$, for the interaction between picture position and sound valence; while $F(1, 44) = 1.76$, $p = .191$, $\eta_p^2 = .038$ for the interaction of sound position and sound valence (all other Fs < 1, ps > .422). In summary, as can also be seen in Figure 2, our results on the BS′ of the neutral–neutral condition closely mirror those of the BS of the unpleasant–neutral condition, with the expected exception of a positive overall BS with valence-asymmetric picture pairs.

In the visual-only block, we conducted the corresponding one-way ANOVA on the BS′ of the first fixations with the position of the A-set picture (left vs. right) as a repeated measures factor. As expected when using arbitrary picture sets, the constant effect was not significant, $F(1, 44) = 0.36$, $p = .551$, $\eta_p^2 = .008$ ($d_Z = 0.09$; $M = 0.01$, $SD = 0.15$, for the average BS′). Consistent with the audio-visual block, there was a significant effect of picture position, $F(1, 44) = 16.73$, $p < .001$, $\eta_p^2 = .275$ ($d_Z = 0.61$), with higher BS′ when the A-set picture was presented on the left ($M = 0.34$, $SD = 0.52$) than on the right ($M = -0.31$, $SD = 0.58$).

A Comparison: Unpleasant–Neutral Versus Neutral–Neutral Picture Pairs

As argued in the introduction, a valence-related interpretation of the two main findings reported above regarding the variable factors (i.e., the effect of picture and sound position) can be potentially maintained by hypothesizing that the result regarding BS is stronger than the one for BS′. Therefore, we conducted the following analyses.

In the audio-visual block, the difference between the BS for the left and the right positions of the reference picture is $M = 0.77$ ($SD = 0.96$); for BS′, this difference is $M = 0.78$ ($SD = 0.96$). Thus, the slight numerical difference is even larger for BS′ than for BS, but $|t| < 1$, $BF_{0+} = 10.55$, $RR_{B>10}$ [0.95, >2],[9] indicating strong evidence for H_0. Furthermore, the difference between BS for sound location congruence versus incongruence with the reference image is

[8] The (expected) exception concerns Gerdes et al.'s (2021) finding of a slightly increased left–right difference in the BS when the sound was unpleasant: Our results did not show this interaction, $F(1, 44) = 0.86$, $p = .360$, $\eta_p^2 = .019$. Note, however, that the power was only $1 - \beta = .48$ (see also the introduction with regard to this effect). Descriptively, BS for the left picture position was $M = 0.43$, $SD = 0.48$ and $M = 0.46$, $SD = 0.05$ in the unpleasant and neutral sound conditions, respectively; BS for the right picture position was $M = -0.32$, $SD = 0.55$ and $M = -0.33$, $SD = 0.53$ in the unpleasant and neutral sound conditions, respectively.

[9] When the criterion value is just passed (as with >10 in the present case), the corresponding RR is typically very small. Thus, it is more informative to examine whether the next lower criterion demonstrates robustness. In the present case, it is $RR_{B>3}$ [0.24, >2] for the criterion value of >3.

$M = 0.08$ ($SD = 0.26$), whereas this difference is $M = 0.13$ ($SD = 0.25$) for BS′. Thus, numerically, the difference for BS′ is greater than for BS, but it is associated with $|t| < 1.19$, $BF_{0+} = 17.65$, $RR_{B>10}$ [0.56, >2], reflecting strong evidence in favor of the H_0.

In the visual-only block, both the difference between the left and right BS and the difference between the left and right BS′ are exactly $M = 0.65$ (SDs = 1.13 and 1.06, respectively).

Subsequent Dwell Time

Unpleasant–Neutral Picture Pairs

As secondary analyses, we performed the same analyses on the BS/BS′ of the subsequent dwell time as the dependent variable as for the BS/BS′ of the first fixations. In the audio-visual block of the unpleasant–neutral condition, the analogous repeated measures ANOVA revealed a significant constant effect, $F(1, 44) = 8.49$, $p = .006$, $\eta_p^2 = .162$, with a positive BS across conditions ($M = 0.09$, $SD = 0.21$), replicating the bias of sustained attention toward unpleasant pictures. In contrast to the BS of initial attention, but consistent with the results of Gerdes et al. (2021), picture position had no significant effect on the distribution of subsequent attention, $F(1, 44) = 0.79$, $p = .380$, $\eta_p^2 = .018$ (BS for the left picture position: $M = 0.07$, $SD = 0.25$; BS for the right picture position: $M = 0.12$, $SD = 0.31$). In accordance with the results of first fixations, the effect of sound position was significant, $F(1, 44) = 4.44$, $p = .041$, $\eta_p^2 = .092$, with an increased BS for the same ($M = 0.12$, $SD = 0.27$) compared to the opposite sound position relative to the reference picture ($M = 0.06$, $SD = 0.18$). The main effect of sound valence (see Gerdes et al., 2021) was not replicated in the present study, $F(1, 44) = 0.43$, $p = .516$, $\eta_p^2 = .010$ (BS with unpleasant sounds: $M = 0.10$, $SD = 0.24$; BS with neutral sounds: $M = 0.08$, $SD = 0.21$); all other Fs < 1.68, ps $> .202$.

In the visual-only block, the corresponding one-way ANOVA on the BS of dwell time revealed a significant constant effect, $F(1, 44) = 7.41$, $p = .009$, $\eta_p^2 = .144$, with a positive BS ($M = 0.10$, $SD = 0.24$). Consistent with the results of the audio-visual block, the BS did not differ between the left ($M = 0.06$, $SD = 0.32$) and the right ($M = 0.13$, $SD = 0.36$) picture position, $F(1, 44) = 0.91$, $p = .345$, $\eta_p^2 = .020$.

Neutral–Neutral Picture Pairs

We performed the analogous analyses in the neutral–neutral condition on the BS′ of dwell time as on the BS′ of first fixations. In the audio-visual block, the corresponding repeated measures ANOVA, as expected, did not yield a significant constant effect, $F(1, 44) = 0.09$, $p = .771$, $\eta_p^2 = .002$ ($M = 0.00$, $SD = 0.07$ for the BS′ of dwell time across conditions). Consistent with the results for unpleasant–neutral pictures, there was a significant main effect of sound position, $F(1, 44) = 14.27$, $p < .001$, $\eta_p^2 = .245$. Again, when the sound was presented on the same side as the reference picture, there was a higher BS′ ($M = 0.05$, $SD = 0.11$) compared to the opposite side ($M = -0.05$, $SD = 0.11$). Similar to the results of the unpleasant–neutral condition, BS′ did not differ between the left ($M = -0.03$, $SD = 0.17$) and right ($M = 0.04$, $SD = 0.20$) presentation of the reference picture, $F(1, 44) = 1.97$, $p = .168$, $\eta_p^2 = .043$. There was also no significant effect of sound valence, $F(1, 44) = 1.70$, $p = .199$, $\eta_p^2 = .037$ (BS′ with unpleasant sounds: $M = 0.02$, $SD = 0.08$; BS′ with neutral sounds: $M = -0.01$, $SD = 0.10$). No further effect reached significance: The interaction between sound position and sound valence was associated with $F(1, 44) = 3.65$, $p = .062$, $\eta_p^2 = .077$; the interaction between picture position and sound valence was associated with $F(1, 44) = 1.46$, $p = .233$, $\eta_p^2 = .032$; for all other effects, Fs < 1, ps $> .790$. In sum, the effects of the variable factors on the BS′ of dwell time closely mirrored the results on the BS with unpleasant–neutral picture pairs.

In the visual-only block, as expected, the corresponding one-way ANOVA did not show a significant constant effect, $F(1, 44) = 0.06$, $p = .938$, $\eta_p^2 < .001$ ($M = 0.00$, $SD = 0.09$ for the average BS′). Consistent with the audio-visual block, no picture position effect emerged, $F(1, 44) = 1.79$, $p = .187$, $\eta_p^2 = .039$ (BS′ for the left position: $M = -0.05$, $SD = 0.27$; BS′ for the right position: $M = 0.05$, $SD = 0.25$).

Discussion

A growing body of empirical evidence indicates that visual spatial attention is preferentially directed to emotional over neutral stimuli, often referred to as an attentional bias for emotional information (e.g., Yiend, 2010). The important question of whether this attentional bias is boosted by other factors, such as left versus right hemifield presentation or converging cross-modal information (Alpers, 2008; Gerdes et al., 2021), was tested as differences in attentional BS. Specifically, Gerdes et al. (2021) investigated these questions by calculating a BS of initial and sustained attention in favor of the unpleasant picture when presenting unpleasant and neutral pictures in pairs. Focusing on the initial allocation of attention, the BS was defined as the number of first fixations on the unpleasant picture minus the number of first fixations on the neutral picture in a given condition, divided by the total number of trials in the condition.

As argued in the introduction, we agree that the average of this dependent variable can be unequivocally interpreted as an attentional bias to unpleasant pictures. However, Gerdes et al. (2021) interpreted the moderation of this BS by the variable features of the picture presentation as evidence for a moderation of an emotional attention bias. In contrast, we argued that the observed moderations of the BS are not related to the valence of the pictures that make up the picture pairs. To corroborate our arguments, we conducted an extended replication of Gerdes et al.'s study. We directly tested whether the effects of the variable factors on the BS occur specifically in a setting in which unpleasant and neutral pairs of pictures are presented. In contrast, if comparable results also appear with valence-invariant (i.e., neutral–neutral) picture pairs, this would indicate that these effects do not depend on the valence combination of the picture pairs. In sum, our main goal was to test whether the effects of the variable factors occur on a corresponding attentional BS when neutral picture pairs are presented, that is, on a BS for an arbitrary set of neutral pictures.

First of all, the present study showed a pattern highly comparable to that of Gerdes et al. (2021) for unpleasant–neutral pairs of pictures. Foremost, we have replicated the bias of initial attention toward unpleasant pictures when paired with neutral ones, as indicated by the significant positive BS. This finding was further supported by strong Bayesian evidence for the hypothesis of a positive BS. Thus, participants' initial attention was more often allocated to the unpleasant than to the neutral picture, confirming previous findings (e.g., Alpers, 2008; Gerdes et al., 2021; for reviews, see, e.g., Pourtois et al., 2013; Yiend, 2010). We then tested whether the factors of interest, the variable features of the picture presentation, moderated the BS: the position of the unpleasant picture and the position of the accompanying sound (see Gerdes et al., 2021). Indeed, we replicated both effects: BS was enhanced when the unpleasant picture appeared on the left (as opposed to the right) and when the sound location corresponded to that of the unpleasant picture.

Critically, however, the same pattern of results emerged in the condition in which only neutral pictures were presented. That is, we computed a BS′ of initial attention with respect to an arbitrarily chosen, counterbalanced set of neutral pictures (the A-set pictures), and this BS′ was also moderated by the variable features of the picture presentation. Specifically, BS′ differed by picture position (here, referring to the position of the A-set picture), indicating that the BS was enhanced when the A-set picture was presented on the left side compared to the right side. The similarity in the pattern of results across picture pair conditions was further supported by extreme Bayesian evidence in favor of a leftward bias in both conditions. Mirroring the results of the unpleasant–neutral condition, a main effect of sound position also emerged: BS′ was enhanced when the sound was presented in the same location as the A-set pictures. In the unpleasant–neutral condition, this effect was associated with moderate Bayesian evidence, whereas in the neutral–neutral condition, it was supported by strong Bayesian evidence.

How can we explain that the arbitrary BS′ shows a leftward bias and cross-modal facilitation? As explained in the introduction, we can interpret both the average BS and the average BS′ as indices of preferential attention to the reference picture set. However, this is nonsensical for BS′, given that the concrete picture set used as a reference in the calculation of BS′ was counterbalanced across participants. In contrast, when interpreting the effects of the variable features of the picture presentation, there is no need to distinguish between the picture sets that constitute the pairs. The observed effects on the BS and BS′ can thus be seen as a main effect of spatial asymmetry of the initial allocation of visual attention and as a main effect of spatial congruence between sound and picture. The latter effect can be interpreted in line with the well-established exogenous spatial cueing effect (e.g., Chica et al., 2014; Posner, 1980; in a cross-modal setting, see, e.g., Lee & Spence, 2015, 2017; and for a review) and, more generally, in line with facilitated visual processing when the location of a sound and a visual stimulus coincide (McDonald et al., 2000, 2013; for a review, see Störmer, 2019). That is, the presentation of a sound facilitates spatial attention to its location, leading to facilitated orienting to and processing of visual stimuli at that location. Regarding the effect of picture position, there is evidence that visual attention can be characterized by a *general* leftward bias (e.g., in scene perception, e.g., Foulsham & Kingstone, 2010; Foulsham et al., 2013), in line with an assumed left hemifield advantage of visual spatial attention (e.g., Siman-Tov et al., 2007), a leftward bias in higher level attentional areas (e.g., Orr & Nicholls, 2005; Ossandón et al., 2014), or a left-to-right scanning bias (e.g., Chokron et al., 1998). These possible interpretations are in no way dependent on the content of the picture sets. Although they are plausible in view of the former literature, our study was not designed to test these alternative interpretations.

We argued in the introduction that our study might give hints for possible differences between the unpleasant–neutral and neutral–neutral conditions with regard to the moderations, for example, whether the picture location effect on BS is larger than the one for BS′. Of course, we did not plan to test these differences with sufficient test power, simply because we implicitly assumed for the planning of the sample size that these differences would be nonexistent. Specifically, we based the expected effect sizes for the variable factors on BS′ on the corresponding effects for BS found by Gerdes et al. (2021). However, in principle, it can be an intriguing question for future research to examine whether the effects demonstrated for both unpleasant–neutral and neutral–neutral conditions are indeed more or less pronounced for unpleasant–neutral than for neutral–neutral pairs. This would indicate influences on a bias that could be interpreted in terms of valence. On a related note, it could be investigated whether such possible differences are specific to unpleasant pictures or extend to positive valence (e.g., Brosch et al., 2008; Pool et al., 2016; see also Folyi & Wentura, 2019) and to classes of emotional stimuli other than scene pictures (e.g., dynamic stimuli). So the questions remain: Is there an increased bias of visual attention toward emotional stimuli when presented in the left hemifield, and is this effect facilitated by emotionally congruent sounds? We argue, however, that these important questions need to be addressed by using an appropriate design, thus by including a neutral–neutral baseline against which BS can be compared.

The starting point for such an endeavor is, of course, sobering in view of our results, since we found almost no differences between the effects for BS and BS′ as dependent variables. Thus, someone who hypothesizes the existence of such differences must assume that our study underestimates them. Given our results, what is a plausible effect size estimate for power planning? Perhaps the assumption of a small effect (i.e., $d_Z = 0.20$) would be the highest acceptable value. Note that such an experiment would require a sample size of $N = 272$ to achieve a power of $1 - \beta = .95$, given $\alpha = .05$ (one-tailed), a rather large number for a single-session eye-tracking laboratory experiment. A better choice might be to conduct a sequential Bayes study (e.g., Schönbrodt et al., 2017); that is, BFs are calculated during recruitment until an a priori defined level of evidence is reached. If the BF reaches a predefined value in favor of H_0 or H_1, sampling is stopped, and the favored hypothesis is accepted. In this regard, the BFs obtained with our relatively small sample are an interesting starting point. First, with regard to the hypothesis that the effect of picture location on BS is larger than the one on BS′, we obtained strong evidence in favor of the H_0. Second, with regard to the hypothesis that the effect of the location congruence between the sound and the reference picture is larger on BS than on BS′, we also obtained strong evidence in favor of the H_0. Thus, the best guess in light of our results is that the moderation of BS by picture location and sound location congruence is not related to the unpleasantness of the pictures.

Taken together, in this article, we argue that far-reaching claims in the field of emotion and attention, such as the existence of a systematic leftward bias in emotional attention and the enhancement of an emotional attention bias by auditory cues, should be tested using methodologically sound approaches. Our findings show that the design used in previous research did not, in fact, provide a valid test of these hypotheses, and we offer specific suggestions for future studies. Importantly, the methodological issue we highlight is not limited to investigations of leftward bias in emotional attention. Similar designs, in which pairs of images with contrasting valence are presented, are widely used to address other theoretical questions in emotional attention. In such cases, analogous interpretational problems can arise when comparable BS and designs are employed. By drawing attention to this issue and offering a solution, we aimed to contribute to improving research design and the validity of interpretations more generally.

Constraints on Generality of the Findings

The participants in our study were students at Saarland University, Germany, and therefore human. As our theoretical considerations and hypotheses relate to general human cognitive processes, this gave us the opportunity to test and possibly falsify our claims. This statement should not be confused with a claim that our results generalize to other human subpopulations.

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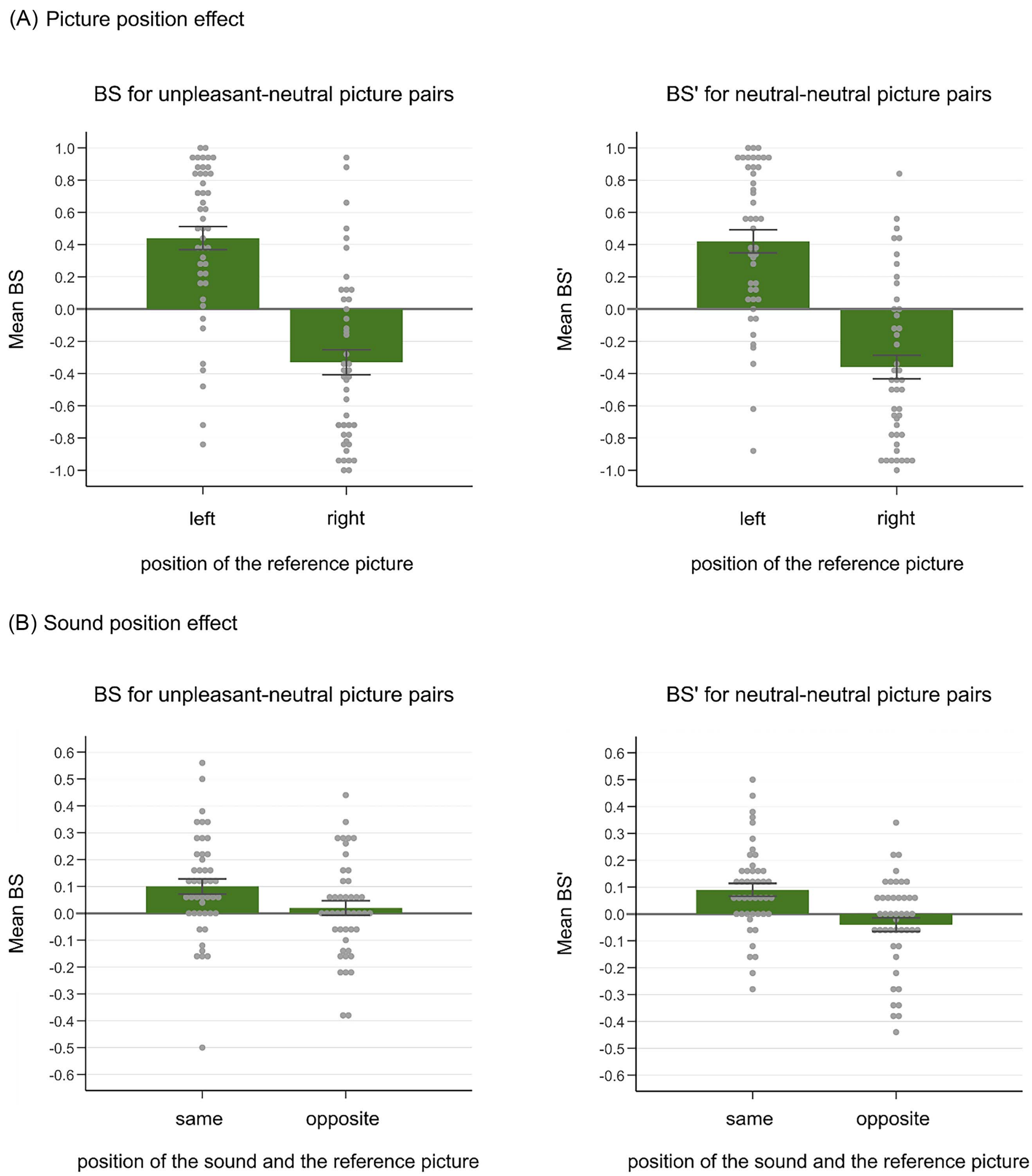
(*Appendix follows*)

Appendix

Supplements to the Results

Figure A1

Mean BS of the First Fixations in the Audio-Visual Block in the Unpleasant–Neutral (BS, Left Column) and Neutral–Neutral Picture Pair Conditions (BS′, Right Column) of the Present Study



Note. The gray dots represent individual data points. (A) Mean BS for the left versus right position of the reference picture (i.e., unpleasant picture for BS, A-set picture for BS′). (B) Mean BS for the same versus opposite sound position relative to the reference picture. Error bars represent the standard error of the means. BS = bias scores. See the online article for the color version of this figure.

(*Appendix continues*)

Results of the Unpleasant–Unpleasant Picture Pair Condition

The unpleasant–unpleasant picture pair condition was introduced into our design to avoid unpleasant pictures being a rare category compared to neutral pictures. Although it was not a central aim of the study to test our hypotheses on a BS′ for another valence-invariant picture pair condition, we report the results of this condition for the sake of completeness.

Number of First Fixations

We performed the analogous analyses in the unpleasant–unpleasant condition as in the neutral–neutral condition. Thus, in the audio-visual block, we conducted a 2 (position of the A-set picture: left vs. right) × 2 (position of the sound: same vs. opposite to the A-set picture) × 2 (sound valence: unpleasant vs. neutral) repeated measures ANOVA on the BS′ of the first fixations. As expected, there was no significant constant effect, $F(1, 44) = 0.01$, $p = .978$, $\eta_p^2 < .001$ (the average BS′ was $M = 0.00$, $SD = 0.10$). As in the other picture pair conditions, there was a significant picture position effect, $F(1, 44) = 32.20$, $p < .001$, $\eta_p^2 = .423$, with larger BS′ for the left ($M = 0.42$, $SD = 0.51$) compared to the right image position ($M = -0.42$, $SD = 0.50$). Consistent with the results of the BS and BS′ for neutral–neutral pictures, there was a significant effect of sound position, $F(1, 44) = 17.18$, $p < .001$, $\eta_p^2 = .281$, with larger BS′ for the same ($M = 0.07$, $SD = 0.17$) compared to the opposite sound position ($M = -0.07$, $SD = 0.13$) as the reference picture. Unexpectedly, a significant main effect of sound valence was also observed, $F(1, 44) = 5.39$, $p = .025$, $\eta_p^2 = .109$, with larger BS′ when accompanied by neutral sounds ($M = 0.03$, $SD = 0.13$) compared to unpleasant sounds ($M = -0.03$, $SD = 0.12$). No further effects were significant, $F(1, 44) = 0.27$, $p = .603$, $\eta_p^2 = .006$, for the interaction of picture position and sound valence (all other Fs < 1, ps $> .435$).

In the visual-only block, we conducted the corresponding one-way ANOVA with the position of the A-set picture as the repeated measures factor. Again, the constant effect was not significant, $F(1, 44) = 0.05$, $p = .818$, $\eta_p^2 = .001$ ($M = 0.01$, $SD = 0.16$, for the average BS′). Consistent with the audio-visual part, the effect of picture position was significant, $F(1, 44) = 21.53$, $p < .001$, $\eta_p^2 = .329$, with higher BS′ for left ($M = 0.37$, $SD = 0.57$) compared to right ($M = -0.36$, $SD = 0.53$) picture position.

Subsequent Dwell Time

In the audio-visual block, as expected, the analogous repeated measures ANOVA on the BS′ of dwell time with the factors picture position, sound position, and sound valence did not yield a significant constant effect, $F(1, 44) = 1.67$, $p = .286$, $\eta_p^2 = .062$ ($M = -0.01$, $SD = 0.06$, for the BS′ of dwell time across conditions). As in the other picture pair conditions, there was no significant effect of picture position, $F(1, 44) = 0.02$, $p = .891$, $\eta_p^2 < .001$ (BS′ for left picture position: $M = -0.01$, $SD = 0.20$; for right picture position: $M = -0.01$, $SD = 0.18$). On the BS′ of the dwell time, the effect of sound position did not reach significance, $F(1, 44) = 3.60$, $p = .064$, $\eta_p^2 = .076$ (BS′ for the same sound position: $M = 0.02$, $SD = 0.11$; for the opposite sound position: $M = -0.04$, $SD = 0.11$). No further effects reached significance; the interaction of picture position and sound valence was associated with $F(1, 44) = 3.16$, $p = .082$, $\eta_p^2 = .067$; all other Fs < 2.01, ps $> .164$.

In the visual-only block, as expected, the one-way ANOVA with the factor picture position did not show a significant constant effect, $F(1, 44) = 2.05$, $p = .159$, $\eta_p^2 = .045$ ($M = -0.02$, $SD = 0.09$, for the average BS′). Again, no picture position effect emerged, $F(1, 44) = 0.01$, $p = .919$, $\eta_p^2 < .001$ (BS′ for the left position: $M = -0.02$, $SD = 0.23$; BS′ for the right position: $M = -0.02$, $SD = 0.22$).

Discussion

Overall, as expected, the results for the BS′ of the unpleasant–unpleasant condition closely mirrored the results for the BS′ of the neutral–neutral condition. Thus, the effects of picture position and sound position on the BS′ of the first fixations were replicated in a further valence-invariant picture pair condition. This again supports the conclusion that the effects of the variable features of the picture presentation (i.e., the position of the reference picture, whether it is accompanied by an unpleasant or neutral sound) on the BS are not specific to unpleasant–neutral picture pairs. However, there was also a significant main effect of sound valence on the BS′ of first fixations when presenting unpleasant picture pairs. As this effect reflects increased first fixations when accompanied by neutral (as opposed to unpleasant) sounds to an arbitrary set of unpleasant pictures, whose assignment as reference picture set was counterbalanced across participants, it is open to further discussion whether this effect is due to random fluctuations only. Indeed, while there were no statistical outliers in the BS′ difference between the neutral and unpleasant sound conditions, the median of the BS′ difference between these conditions was exactly $Mdn = 0.00$.

Received January 25, 2025
Revision received September 26, 2025
Accepted September 27, 2025 ■